

Name: _____ Date: _____

Period: _____

No notes. Show enough work that a wrong answer can still earn a right idea.

- 1.
- Solve, then classify:**
- $4(x - 3) = 2(2x - 5) + 2$
- .

Work:

$$4(x - 3) = 2(2x - 5) + 2$$

$$4x - 12 = 4x - 10 + 2$$

$$4x - 12 = 4x - 8$$

$$-12 = -8$$

Circle one: one solution · **no solution** · infinitely many solutionsWhat in your last line told you which? **The variable dropped out and left $-12 = -8$, which is *false* — so no value of x makes the equation true. It is a contradiction.**

- 2.
- Solve by factoring:**
- $x^2 - 3x = 10$
- .

Step 1 — get one side to zero:

$$x^2 - 3x = 10$$

$$x^2 - 3x - 10 = 0$$

Step 2 — factor and apply the zero product property:

$$(x - 5)(x + 2) = 0$$

$$x = 5 \quad \text{or} \quad x = -2$$

Verify one of your solutions in the original equation. **At $x = 5$: $5^2 - 3(5) = 25 - 15 = 10$. (Or at $x = -2$: $4 + 6 = 10$.)**

- 3.
- SOL-style multiple choice.**
- What is the solution set of
- $-3x + 7 > 19$
- ?

$$-3x + 7 > 19$$

$$-3x > 12$$

$$x < -4$$

A. $x > -4$

B. $x < -4$ ← **correct**

C. $x > 4$

D. $x < 4$

In one sentence, what did the *wrong* answer you were most tempted by get wrong? **Answers vary — A keeps the symbol after dividing by a negative, C and D divide by 3 instead of -3 and lose the sign of the endpoint.**

Teacher Note

Sort the tickets into four piles: *got it / classification* (item 1 answered “one solution,” or abandoned mid-page because the x disappeared) / *inequality direction* (item 3 — A is the flip, C and D divide by $+3$) / *zero-first* (item 2 attacked as $x(x - 3) = 10$).

The zero-first pile is the one to act on. “ $x(x - 3) = 10$, so $x = 10$ or $x - 3 = 10$ ” has the zero product property without its precondition — the most common quadratic error on the SOL, and the one that matters most for Lesson 1.3, where every *zero of a function* is found by solving exactly this kind of equation. Re-run item 2 with new numbers at the top of 1.3 if the pile is large.